Protocollo

1. Room preparation
2. Loading
3. Short high-intensity ozone sanitation
4. Continuous low-level ozone maintenance

SECTION 1 — SYSTEM ENTRY POINT

What This Ozone Layer Is — and How to Use It Correctly

Reader: A winery operator who already has a functioning grape dehydration process but still experiences unpredictable microbial spoilage, mold outbreaks, or instability during drying.

Goal of this section:

Get you to **integrate ozone into your existing process correctly by understanding exactly where it fits, what it controls, and what parts of your current system must remain unchanged.**

This product is not a dehydration system

You already have a dehydration process.

That means you already have:

* airflow

* humidity management

* temperature control

* drying timelines

* grape handling procedures

This protocol does not replace those systems.

It adds a controlled ozone layer on top of them.

Your objective is not to redesign drying.

Your objective is to:

* reduce microbial instability during dehydration

* suppress mold pressure during long drying cycles

* stabilize the room biologically without changing your core process

The ozone layer has only two functions

This protocol uses ozone in two distinct ways:

1. High-intensity sanitation (short duration)

A one-time operation performed immediately after grape loading.

Purpose:

* rapidly reduce microbial load in the room and on exposed surfaces

* establish a low-contamination starting condition

This is a reset phase.

2. Low-level maintenance ozone (continuous)

A minimal ozone output maintained throughout the drying cycle.

Purpose:

* suppress microbial regrowth during dehydration

* maintain environmental stability over time

This is not active sanitation.

It is a maintenance layer.

What you are NOT trying to do

Do not use ozone to:

* accelerate dehydration

* compensate for poor airflow

* fix overloaded grape placement

* correct unstable humidity conditions

* rescue already contaminated batches

This protocol only works when:

* your existing dehydration process already functions mechanically

* ozone is used as a microbial control layer, not a correction tool

The 4-step operating sequence

You will apply ozone in the same order every cycle:

Step 1 — Prepare the room

Clean and sanitize the drying environment so ozone can work effectively.

Step 2 — Load the grapes

Place grapes so airflow and ozone exposure remain uniform across the room.

Step 3 — Apply a strong ozone sanitation pulse

Run a short-duration high-intensity ozone cycle immediately after loading.

This establishes the low-microbial baseline.

Step 4 — Maintain minimal ozone during dehydration

Run continuous low-output ozone during the drying cycle to suppress microbial growth over time.

This maintains environmental stability.

Your rule for using this protocol

Once the cycle begins:

* do not improvise
* do not adjust ozone randomly
* do not react to every small variation
* do not change process structure mid-cycle

Consistency is more important than intensity.

Before continuing to Step 1

Confirm the following:

* Your drying process already functions operationally
* You are adding ozone as a microbial stability layer
* You are willing to run the same sequence every batch
* You are not expecting ozone to compensate for mechanical process failures

If all four are true, continue to Section 2.

SECTION 2 — ROOM PREPARATION

Step 1: Prepare the Room (Sanitation Baseline)

Reader: A winery operator using an existing grape dehydration system who is adding an ozone layer to reduce mold and microbial spoilage during drying.

Goal of this section:

Get you to **prepare the drying room so ozone can operate at full effectiveness by removing residual contamination and ensuring uniform exposure conditions before grapes are loaded.**

Do this before every batch

This step is not optional and not “periodic maintenance.”

If the room is not prepared correctly, ozone performance becomes inconsistent and unpredictable.

Step 1: Remove all residual organic material

Before anything else:

* Clear all remaining grapes or dried clusters from previous batches

* Remove loose stems, debris, and organic residue from floors and trays

* Empty any collection points where organic matter accumulates

No visible biological material should remain in the room.

Step 2: Reset all contact surfaces

Every surface that will be exposed to grapes or airflow must be cleared:

* racks

* trays

* walls within the drying zone

* fan housings and airflow channels

The objective is not visual cleanliness—it is removing residual microbial load that can survive between cycles.

Step 3: Verify airflow is unobstructed

Before ozone is introduced:

* Ensure air circulation paths are fully open

* Remove any objects blocking vents or fans

* Confirm airflow moves consistently through all zones identified in your system layout

If airflow is uneven now, it will remain uneven during ozone cycles.

Step 4: Confirm room is in a sealed operational state

Before moving to ozone application:

* Close external access points

* Ensure the room is isolated from uncontrolled air exchange

* Confirm the system is ready to operate as a closed treatment environment during sanitation

You are preparing a controlled chamber, not an open environment.

Step 5: Establish “ready for loading” condition

The room is ready only when:

* No organic residue remains

* All surfaces are cleared

* Airflow is unobstructed and stable

* The environment is sealed for controlled operation

If any of these are not true, do not proceed.

Output you must achieve before moving on

At the end of this step, your drying room must be:

* physically clean of residual organic matter

* structurally clear for airflow and ozone movement

* sealed and ready for controlled treatment

* free of any hidden contamination sources from previous batches

What you do NOT do in this step

* Do not apply ozone yet

* Do not load grapes

* Do not adjust humidity or temperature systems

* Do not begin any drying process operations

Why this step matters (in one line)

Ozone cannot reliably reduce microbial load if it is being constantly reintroduced by residual organic contamination in the room.

SECTION 3 — GRAPE PLACEMENT CONDITIONS

Step 2: Place Grapes in the Room

Reader: A winery operator using an existing grape dehydration system and adding a controlled ozone layer to reduce mold and stabilize drying outcomes.

Goal of this section:

Get you to **load grapes into the drying room in a way that guarantees even ozone exposure and airflow penetration across all clusters, preventing hidden moisture pockets and localized spoilage.**

Do this immediately after Room Preparation (Step 1)

Once the room is clean and sealed, the next variable that determines success is not the environment—it is **how you physically place the grapes inside it.**

Most failures at this stage come from placement, not climate settings.

Step 1: Maintain mandatory spacing between clusters

Every grape cluster must be placed so that:

* it does not touch neighboring clusters

* air can move freely around all sides

* no compressed or overlapping contact points exist

If clusters touch, you create trapped zones where ozone and airflow cannot reach consistently.

Step 2: Use uniform layer depth across all racks

Load grapes in consistent thickness:

* avoid dense piles in any section

* avoid thin and thick zones in the same rack

* keep layer depth identical across the entire room

Uneven depth creates uneven drying and uneven microbial suppression.

Step 3: Align placement with airflow direction

Using the airflow pattern established in your system:

* orient racks so air passes through, not against, the grape mass

* ensure no rack blocks a full airflow path

* avoid perpendicular “barrier loading” against ventilation flow

Airflow must move through grapes, not around them.

Step 4: Eliminate structural dead zones

Do not create areas where air movement is restricted:

* no fully enclosed corners of dense loading

* no tight stacking near walls or corners

* no blocked ventilation channels caused by overloading

If air cannot pass through a zone, ozone cannot stabilize it.

Step 5: Balance load distribution across the room

The total grape load must be evenly distributed:

* do not overload one section and underload another

* avoid clustering all high-density loads in a single area

* maintain consistent spatial density across all zones

Imbalance creates localized spoilage risk even if overall conditions are correct.

Output you must achieve before moving on

Your loaded room must have:

* consistent spacing between all clusters

* uniform load depth across racks

* full airflow accessibility throughout all zones

* no structural dead zones or blocked paths

* even distribution of total grape mass

What you do NOT do in this step

* Do not apply ozone yet

* Do not adjust environmental controls to compensate for poor placement

* Do not begin drying cycle timing

* Do not “fix spacing later” after loading

Why this step matters (in one line)

Even perfectly controlled ozone and environmental conditions cannot compensate for grape placement that blocks airflow and traps moisture at the cluster level.

SECTION 4 — HIGH-INTENSITY OZONE RESET

Step 3: One-Time Strong Sanitation Pulse

Reader: A winery operator running an existing grape dehydration process who is integrating ozone to reduce microbial instability, mold outbreaks, and uneven spoilage during drying.

Goal of this section:

Get you to **run a single high-intensity ozone cycle immediately after loading grapes to establish a low-microbial baseline across the entire drying room before maintenance ozone begins.**

Do this immediately after Step 2 (Grape Placement)

Once grapes are loaded correctly, the system must be “reset” biologically before dehydration continues.

This is the only point in the process where ozone is used at maximum intensity.

Step 1: Seal the room completely

Before starting the cycle:

* Close all access points

* Ensure no external air exchange is occurring

* Confirm airflow system is in recirculation mode (if applicable)

* Verify no human access during the cycle

The room must function as a closed treatment chamber.

Step 2: Activate high-intensity ozone cycle

Run ozone at strong sanitation level with the following constraints:

* full room exposure (air and surfaces)

* uninterrupted cycle duration (do not stop mid-process)

* consistent output level throughout the cycle

The objective is uniform exposure, not localized concentration.

Step 3: Maintain airflow during treatment

During the ozone cycle:

* keep internal circulation active

* ensure air moves through all loaded zones

* avoid stagnant pockets where ozone cannot circulate

Ozone must reach all grape surfaces evenly.

Step 4: Do not interfere during the cycle

Once started:

* do not adjust settings

* do not open the room

* do not interrupt airflow configuration

* do not shorten or extend the cycle mid-run

The cycle must run as a single uninterrupted operation.

Step 5: Transition to post-cycle stabilization

After the cycle completes:

* allow the room to stabilize internally

* prepare for safe air clearance (to be done before Step 4)

* ensure ozone concentration returns to safe operational levels before continued access

Do not begin maintenance ozone until stabilization is complete.

Output you must achieve before moving on

After this step:

* microbial load in air and on surfaces is significantly reduced

* the drying room is reset to a low-contamination baseline

* grapes are now in a controlled starting environment for long-cycle drying

What you do NOT do in this step

* Do not use this cycle repeatedly

* Do not treat this as a continuous process

* Do not load or unload grapes during the cycle

* Do not begin maintenance ozone yet

* Do not adjust environmental conditions in response to cycle behavior

Why this step matters (in one line)

Without a single high-intensity reset after loading, you begin the drying cycle on an already active microbial baseline that maintenance ozone cannot fully compensate for.